Technical Report: Life Improvement Optimization Models

Sepehr Barekati

Introduction

This report outlines the formulation and solution of three optimization problems aimed at life improvement. The models include a dietary optimization framework (Linear Programming), a project management resource allocation model (Linear Programming), and a financial portfolio selection strategy (Quadratic Programming). Solutions were derived programmatically using Python and the `cvxpy` library.

1 Problem 1: Appropriate Diet

The objective of this problem is to minimize the total cost of a daily diet while satisfying minimum nutritional requirements.

1.1 Mathematical Model

Let $N = 10$ be the number of available food items and $M = 5$ be the number of required nutrients. We define x_i as the quantity of food i consumed.

$$\begin{aligned} \text{Minimize} \quad & Z = \sum_{i=1}^{N} C_i x_i \\ \text{Subject to} \quad & \sum_{i=1}^{N} P_{ji} x_i \geq B_j \quad \forall j \in \{1, \ldots, M\} \\ & x_i \geq 0 \quad \forall i \in \{1, \ldots, N\} \end{aligned}$$

Where C_i represents the cost per unit of food i, P_{ji} represents the amount of nutrient j in food i, and B_j is the daily requirement for nutrient j.

1.2 Results

The linear programming model was successfully solved optimally. The minimum total cost of the diet was determined to be **186.16**. The optimal consumption vector indicates that primarily food items 1, 3, 9, and 10 should be selected to meet all physiological requirements at the minimal financial cost.

2 Problem 2: Project Management

This problem models the allocation of resources to project activities such that the total cost is minimized while satisfying constraints on total time, available resources, and minimum expected profit.

2.1 Mathematical Model

Let $N = 5$ be the number of project activities, and x_i be the amount of resources allocated to activity i.

$$\begin{aligned}
\text{Minimize} \quad & Z = \sum_{i=1}^{N} C_i x_i \\
\text{Subject to} \quad & \sum_{i=1}^{N} T_i x_i \leq t \quad \text{(Total Time Limit)} \\
& \sum_{i=1}^{N} x_i \leq R \quad \text{(Total Resource Limit)} \\
& \sum_{i=1}^{N} p_i x_i \geq P \quad \text{(Minimum Profit Target)} \\
& x_i \geq 0 \quad \forall i \in \{1, \dots, N\}
\end{aligned}$$

Here, C_i is the cost per resource unit for activity i, T_i is the time factor, p_i is the profit factor, $t = 800$, $R = 200$, and $P = 2500$.

2.2 Results

The optimal total cost to achieve the project constraints is **50.0**. Due to the high profitability of activity 1 ($p_1 = 100$), the optimizer fully allocated 25 units to activity 1, which strictly satisfies the total profit constraint of 2500 ($25 \times 100 \geq 2500$) and leaves the other activities with a 0 allocation.

3 Problem 3: Portfolio Selection

The final model seeks to construct an optimal investment portfolio by maximizing expected returns while minimizing associated risk, formalized via the Markowitz Portfolio Optimization approach.

3.1 Mathematical Model

Let $n = 10$ be the number of available financial assets. The decision variable x_i represents the fractional weight of total capital invested in asset i.

$$\begin{aligned}
\text{Maximize} \quad & Z = \sum_{i=1}^{n} \mu_i x_i - \lambda \sum_{i=1}^{n} \sum_{j=1}^{n} x_i x_j \sigma_{ij} \\
\text{Subject to} \quad & \sum_{i=1}^{n} x_i = 1 \\
& 0 \leq x_i \leq 1 \quad \forall i \in \{1, \ldots, n\}
\end{aligned}$$

Where μ_i is the expected return of asset i, σ_{ij} is the covariance between assets i and j, and $\lambda = 0.5$ denotes the risk aversion parameter.

3.2 Results

The quadratic program converged to an optimal status. The maximized expected return is **19.68**, with a minimized risk variance of **2.89**. The portfolio distribution strictly weights two distinct assets:

- Asset 4 (x_4): 61.88%
- Asset 9 (x_9): 38.13%

All other available assets received a 0% weight, demonstrating an efficient frontier concentration to mitigate variance while maximizing yield.